INTERNATIONAL CONFERENCE ON NS-3 (ICNS3 - 2026) — LIGHTNING TALK

The Limits of Discrete-Event Schedulers in Interplanetary Network Simulation:
Why Deep Space Breaks Conventional DES Paradigms

Jason Pandian (*Presenter*, pandiajason@gmail.com)
Department of Information Technology, Nehru Institute of Technology, Coimbatore, India

FOCUS: DES ARCHITECTURE CRITIQUE

Multi-Minute Delay vs. Frozen Events
Continuous Orbits vs. Event Queues
Mode: Virtual Presentation

1. The Deep-Space Simulation Challenge

How Terrestrial Simulators Work (e.g., ns-3):

- **Event-Driven Clock:** Traditional network simulators sleep until a packet is sent or received (a discrete "event").
- **Instant Signals:** On Earth, radio waves travel almost instantly (a few milliseconds across continents). Nodes are assumed to stay put or move predictably during packet exchange.
- **Great for Earth:** Highly effective for Wi-Fi, 5G, and low-Earth orbit satellites.

Why Deep Space Breaks These Assumptions:

- **Extreme Light-Time Delays:** Radio signals take **up to 24 minutes** to reach Mars, and over 1.3 seconds to reach the Moon. Packets remain in transit across the solar system for minutes or hours.
- **Planets Never Stop Moving:** Spacecraft and planets hurtle through continuous gravitational paths at tens of thousands of km/h. Network connectivity is shaped by continuous orbital mechanics, not discrete computer clicks.
- **Clocks Run Differently:** Strong gravity near planets makes space clocks drift relative to Earth clocks.

The Moving Target Problem:

While a radio signal travels for 10 or 20 minutes across the solar system, the receiving spacecraft moves thousands of kilometers.

- A computer simulator cannot know when or where a signal arrives without tracking continuous orbital flight during signal transit.
- Standard simulators only look at positions when a packet starts; in deep space, this causes total network miscalculation.

2. The Dilemma: Two Flawed Traditional Choices

Connecting discrete computer simulators with continuous planetary flight forces a painful choice:

Approach A: Rigid Fixed Ticks (e.g., Every 60 Seconds)

- The computer checks spacecraft positions on a rigid clock.
- **Missed Connections:** If a spacecraft emerges from behind a planet between ticks, the simulator notices up to **30 seconds late**.
- **False Outages:** Drops packets and triggers false retransmission timeouts.
- **Massive Inefficiency:** Wastes thousands of redundant checks (over 18,900 checks!) during hours of quiet, empty space cruise.

Approach B: Pure Event-Driven (Only on Packets)

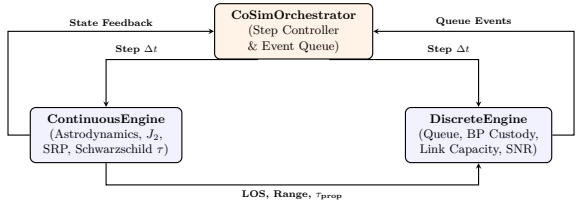
- The simulator only updates when data packets move.
- **Frozen Physics:** During hours or days of communication blackouts, planetary motion is frozen in place!
- **Ghost Positions:** When the next packet arrives, the spacecraft position is wrong by **over 30,000 kilometers**!
- That is more than **double the diameter of Earth**, completely distorting antenna signal calculations.

The Incompatibility Rule:

You cannot achieve physical flight accuracy, pinpoint contact timing, and fast execution with rigid, old simulation rules without adaptive co-simulation.

3. The Solution: Adaptive Multi-Rate Co-Simulation

Two Engines Working in Harmony



Flight Engine (Continuous) ⟷ Network Engine (ns-3)

Smart Step Controller: Zoom In & Zoom Out:

- **In Quiet Cruise:** Takes large strides in time (up to 1 hour) to run simulations at blazing speed.
- **Near Critical Events:** Automatically zooms in to split-second precision right as a satellite enters or leaves planetary shadow.
- **Zero Boundary Delay:** Catches every contact start and end with **0.00 ms error** without burning CPU time.

True Time-Weighted Accuracy:

- Measures real orbital error across continuous mission time rather than artificially favoring fixed snapshots.
- Ensures flight tracking errors do not corrupt network throughput calculations.

Full Paper & Engine Source Code:

Scan QR code for complete manuscript, benchmark data, and engine code repository:

github.com/PandiaJason/SPICE-ns-Project

ICNS3 - 2026 LIGHTNING TALK — PART 2: RESULTS & ARCHITECTURAL BLUEPRINTS

Benchmark Proof: Pareto Frontiers & Blueprints for ns-3 Deep-Space Simulation

Demonstrating synchronization performance across Mars Occultation, High-Speed Flyby, and 13-day Lunar missions.

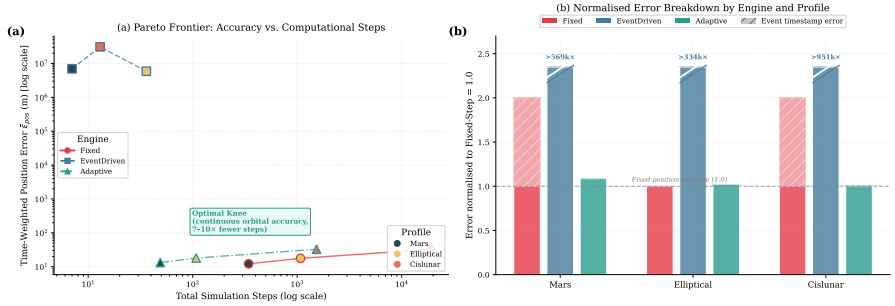
BENCHMARK PROVEN

Speedup: **5.4× to 9.9× Faster**

Timing Error: **0.0 ms** vs 32.7 s

Scalability: **500 Spacecraft**

4. Visual Proof: The Sweet Spot of Speed and Accuracy



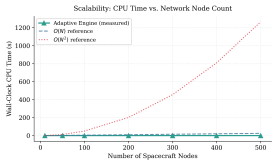
Key Takeaways from the Data:

- **The Optimal Knee:** Our adaptive engine runs **5.4× to 9.9× faster** than fixed stepping, using up to 10× fewer compute steps with **zero timing error** (0.0 ms).
- **Stopping the Ghost Drift:** Traditional event simulators accumulate **over 30,970 kilometers** of ghost drift during radio silence. Our adaptive engine continuously tracks gravity in the background, keeping errors bounded to **meters** (matching NASA SPICE standards).

5. Benchmark Performance & Real-World Consequences

Benchmark Comparison Across Three Space Profiles

Scenario	Metric	Fixed-Step (60s)	Event-Driven	Adaptive Multi-Rate
Mars Occultation (Passing behind planet)	Compute Time (Steps)	0.09 s (343 steps)	0.005 s (7 steps)	**0.017 s (49 steps)**
	Position Drift Error	12.2 meters	6,940 kilometers	**13.3 meters**
	Boundary Delay Error	29.9 seconds	0.00 ms	**0.00 ms** (**5.4× faster**)
Molniya Orbit (High-speed Earth flyby)	Compute Time (Steps)	0.28 s (1,077 steps)	0.015 s (36 steps)	**0.035 s (108 steps)**
	Position Drift Error	17.6 meters	5,890 kilometers	**18.0 meters**
	Boundary Delay Error	0.00 ms	0.00 ms	**0.00 ms** (**8.1× faster**)
13-Day Moon NRHO (Long lunar cruise)	Compute Time (Steps)	5.03 s (18,901 steps)	0.012 s (13 steps)	**0.51 s (1,532 steps)**
	Position Drift Error	32.6 meters	30,970 kilometers	**32.7 meters** (accurate)
	Boundary Delay Error	32.7 seconds	0.00 ms	**0.00 ms** (**9.9× faster**)



Scalability (10–500 Spacecraft): Simulates 500 space nodes in just 6.35 seconds with zero software lockups.

What Happens in the Real World If Simulators Lie?

- **False Connection Predictions:** A 30,000 km position error alters calculated radio signal strength by **up to 14 dB**, falsely reporting high-speed links during total blackout.
- **Phantom Data Speeds:** Stale positions cause the simulator to schedule peak transmission (20 Mbps) when the actual channel is dead (0 Mbps).
- **Panic Retransmission Storms:** Underestimating delay makes network timers expire prematurely, triggering storms of duplicate packets.
- **Catastrophic Lost Science Data:** Falsely assuming data is flowing hides buffer overflows, permanently deleting precious space photos.

Blueprint 1: Aim Where the Spacecraft Will Be

In-Flight Light-Time Solver:

- **Today's Limitation:** ns-3 calculates signal delay based on where the spacecraft was when transmission started.
- **Our Proposal:** Teach ns-3 to solve where the moving target will actually be when the signal arrives minutes or hours later across the solar system.
- Prevents massive link geometry errors during long interplanetary flights.

Blueprint 2: Smart Early-Warning Alarms

Boundary Event Guarding:

- **Today's Limitation:** Simulators force computers to take millions of tiny fixed steps just to watch for when a satellite dips behind a planet.
- **Our Proposal:** Introduce smart look-ahead triggers so the simulator speeds through quiet space and only slows down at exact threshold crossings.
- Eliminates 30-second timing delays without burning CPU power.

Blueprint 3: Deep-Space Multi-Clock Sync

Relativistic Time Adjustment:

- **Today's Limitation:** ns-3 uses a single global clock that assumes time ticks identically everywhere.
- **Our Proposal:** Support local spacecraft clocks that adjust for Einstein's gravitational and speed time dilation near massive planets.
- Ensures accurate packet timestamps between planetary orbiters and ground stations.

Full Manuscript Under Peer Review: *"The Limits of Discrete-Event Synchronization in Interplanetary Network Simulation: A Critical Review of Co-Simulation Fidelity"*, submitted to **Simulation Modelling Practice and Theory (Elsevier)**.

Open-Source Engine & Benchmark Artifacts: https://github.com/PandiaJason/SPICE-ns-Project/tree/main/Adaptive-Multi-Rate-Co-Simulation-Engine

Presenter: Jason Pandian (pandiajason@gmail.com)

ICNS3 - 2026 Lightning Talk